DIGITAL ASSET SETTLEMENT ENGINEERING SERIES

Digital Asset Settlement for Zelle

Executive Architecture Brief for Stablecoin and Cross-Border Settlement

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Audience	Zelle hiring manager and Principal Engineer interview panel Engineering directors, senior architects, and technical executives

Recommendation. Extend the participant-bank model with an invisible, evidence-gated settlement rail. Keep the internal ledger authoritative, Java/Spring in control, chain-specific technology behind adapters, and the first pilot to one asset, one network, one corridor, no consumer wallet, and no bridge.

Evidence and scope notice. Zelle is used only as a public case study. This brief does not describe or infer confidential Early Warning Services architecture, vendors, controls, data, service levels, production plans, or technology selections. It visibly distinguishes public facts, source-backed inference, proposed architecture, organization-profile recommendations, assumptions, and unknowns.

Abstract

This brief proposes how a regulated digital-asset settlement platform could extend a bank-distributed payment experience without turning the customer journey into a crypto product. Zelle's publicly described model—a recognizable service delivered through participating financial-institution applications—makes it a useful case study for separating customer experience from settlement infrastructure. The architecture retains bank channels, aliases, fraud controls, compliance, support, and disputes; adds a Java/Spring control plane, authoritative double-entry ledger, durable payment intent, treasury, controlled signing, network adapters, independent observation, and reconciliation; and treats blockchain as one external leg rather than the source of business truth.

The immediate recommendation is deliberately narrow: discover the corridor and legal authorities, simulate the ledger and state machine, then prove one regulated asset on one approved public network for a limited participant cohort. The architecture does not select Ethereum, Base, Solana, a custody provider, or an issuer on Zelle's behalf. Each remains an evidence-gated decision. The result is a route that can be stopped, reconciled, and exited before expansion—and an engineering foundation that can later support additional corridors, assets, or networks without changing the customer proposition or the core financial invariants.

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Executive thesis. Preserve the bank-owned customer experience and regulated control plane; use a digital asset only as a bounded, reconciled settlement leg. Prove one route before expanding any dimension.

1. The problem: a faster leg is not a complete payment

PUBLIC FACT The global cross-border agenda still targets four persistent outcomes: lower cost, greater speed, better transparency, and broader access. In official remarks, the FSB Deputy Secretary General points to poor data quality and limited standardization as processing frictions, while World Bank corridor data shows that consumer remittance cost remains material on average. The remarks expressly represent the speaker's views rather than necessarily those of the FSB or its members. (Moloney 2026; World Bank 2025) Those are end-to-end problems. They are not solved merely by making a token transfer fast.

A conventional cross-border payment can pass through customer authentication, sanctions and fraud decisions, funding, foreign exchange, correspondent or payout networks, recipient validation, liquidity, accounting, exceptions, support, and legal discharge. Each step has a different authority, clock, failure mode, and repair mechanism. Delay is often the accumulated result of prefunding, cutoffs, time zones, fragmented data, manual exception handling, and uncertainty about who bears a loss. A new settlement leg can improve some of those mechanics, but it can also introduce issuer, chain, custody, privacy, and provider risks.

INFERENCE A regulated stablecoin can offer a programmable, continuously available transfer instrument on a shared network. It may reduce the time between controlled release and recipient-side availability, make a transaction independently observable, and support more direct treasury movement. It does not establish that the sender was entitled to pay, the recipient may lawfully receive, reserves are available, a bank account was credited, a customer promise was fulfilled, or an obligation was legally discharged. FATF's current work also emphasizes that stablecoin reach and interoperability create real illicit-finance and cross-chain control obligations. (Financial Action Task Force 2026)

Architecture thesis. Blockchain can make one external settlement event more direct and observable. The regulated platform must still own intent, policy, accounting, finality judgments, customer status, reconciliation, repair, and loss allocation.

The most dangerous design shortcut is therefore a single state called "settled." A chain can finalize while a payout fails. A customer can see completion while accounting remains open. An accounting period can close while legal ownership is disputed. A provider can report success while an independent node disagrees. Collapsing these conditions makes happy paths look simple by moving ambiguity into incidents, support queues, and financial breaks.

The opportunity is not to replace the payment network with a blockchain. It is to add an explicitly bounded settlement capability behind the network's business controls. The success measure is not transactions per second in isolation. It is whether a defined corridor can complete with correct balances, known authority, governed keys, explicit finality, recoverable ambiguity, timely reconciliation, and a credible exit path at an acceptable cost.

The decision question

Can a participant-facing route produce better end-to-end cost, time, transparency, or liquidity outcomes while preserving—and proving—the controls expected of a regulated payment network? If the answer cannot be demonstrated for one bounded corridor, expansion is premature.

2. The opportunity for Zelle

PUBLIC FACT Zelle publicly describes a network that connects financial institutions and is experienced through the mobile banking applications of participant banks and credit unions. Its current partner materials report availability through more than 2,400 banking and credit-union applications and more than 143 million consumers; the page identifies its institutional survey as Q1 2026. (Zelle 2026) These facts establish distribution and participant context; they do not establish internal architecture, vendors, or a cross-border plan.

That distribution model is strategically important because a settlement innovation need not require a new consumer destination. A bank can remain the customer channel and account relationship. Existing alias patterns, authentication, fraud controls, customer communication, support, and dispute processes can remain at the experience boundary. A digital asset can operate behind that boundary between approved institutional roles, provided the end-to-end obligation and customer treatment are explicit.

PROFILE DECISION The proposed Zelle-profile posture is therefore participant-facing and blockchain-invisible. A customer initiates a familiar business payment through a participating bank. The network creates a durable payment intent, applies policy, reserves liquidity, and coordinates a settlement route. Participants and operators receive business status, not wallet seed phrases, gas prompts, chain addresses, or arbitrary contract access. The recipient receives value through an approved payout or account-credit mechanism. Exceptions remain in governed support and operations workflows.

This creates three forms of optionality:

- **Product optionality.** The customer proposition can remain stable while the network evaluates a new rail. A failed pilot does not strand customers with tokens or keys.
- **Network optionality.** The enterprise contract can support Ethereum, Base, Solana, or another approved network behind adapters without making one chain's SDK the platform domain model.
- **Strategic optionality.** Evidence from actual route economics, control performance, and operational failure can inform later decisions about additional assets, networks, issuer relationships, or an organization-specific L2.

The participant model also clarifies who must own each decision. Banks own customer authentication and account treatment under the agreed scheme. The settlement platform owns intent, orchestration, ledger, policy coordination, participant status, and reconciliation. Treasury owns liquidity and asset exposure. Security and custody controls own signing authority. The selected network owns its consensus rules. Issuers, payout partners, and providers own bounded external responsibilities. Legal agreements define discharge and loss allocation. No one system inherits every authority simply because it has the fastest clock.

What the case study does not assert

The public facts do not show that Zelle currently uses a blockchain, has selected Solana or another network, issues a token, operates a custody system, or plans a specific corridor. Those remain explicit unknowns and profile decision slots. The brief answers what should be built and validated if the business chooses to explore the opportunity.

3. Proposed architecture: control the payment, contain the rail

PROPOSED The architecture divides responsibility into four layers: customer and participant channels; the regulated control plane; bounded settlement capabilities; and independent evidence, reconciliation, and operations. This is a proposed target architecture derived from Volume I, not a claim about an existing Zelle system.

Evidence status: PROPOSED proposed architecture. Component names and responsibilities derive from Volume I and accepted ADRs; no box represents a confirmed EWS deployment.

3.1. Enterprise control plane

Participant APIs accept approved business requests such as create payment, reserve liquidity, release settlement, and query status. Every request becomes a durable payment intent before an external side effect. Policy coordination binds identity, sanctions/fraud disposition, limits, asset, network, corridor, signer policy, and approvals to that intent. The state machine records operations and immutable attempt lineage. A transactional outbox and durable consumer deduplication aim for exactly-once business effects without claiming impossible universal exactly-once delivery.

PROPOSED The internal double-entry ledger remains authoritative for obligations, reservations, positions, fees, suspense, corrections, and close. A chain receipt, custodian statement, issuer report, bank file, or payout confirmation is independent evidence to reconcile—never a replacement for the complete platform record. Corrections are linked reversals or adjustments, not destructive edits. See **ADR-001** and **Volume I: Ledger and State**.

3.2. Bounded settlement capabilities

Treasury and liquidity services allocate prefunded positions and gas/fee capacity. Custody or an HSM/MPC signing boundary evaluates a narrowly scoped signing request; applications never receive raw production keys. Chain adapters translate a business operation into native transaction semantics while retaining nonce, blockhash, fee, contract/program, and attempt evidence. Issuer, FX, bank, and payout integrations use the same bounded gateway pattern.

Managed infrastructure may submit transactions, but a materially independent read path observes inclusion, canonicality, token state, and relevant events. Submission success is not observation truth. A timeout after possible acceptance becomes an ambiguous outcome; the platform investigates the original identity rather than creating a new value-moving intent.

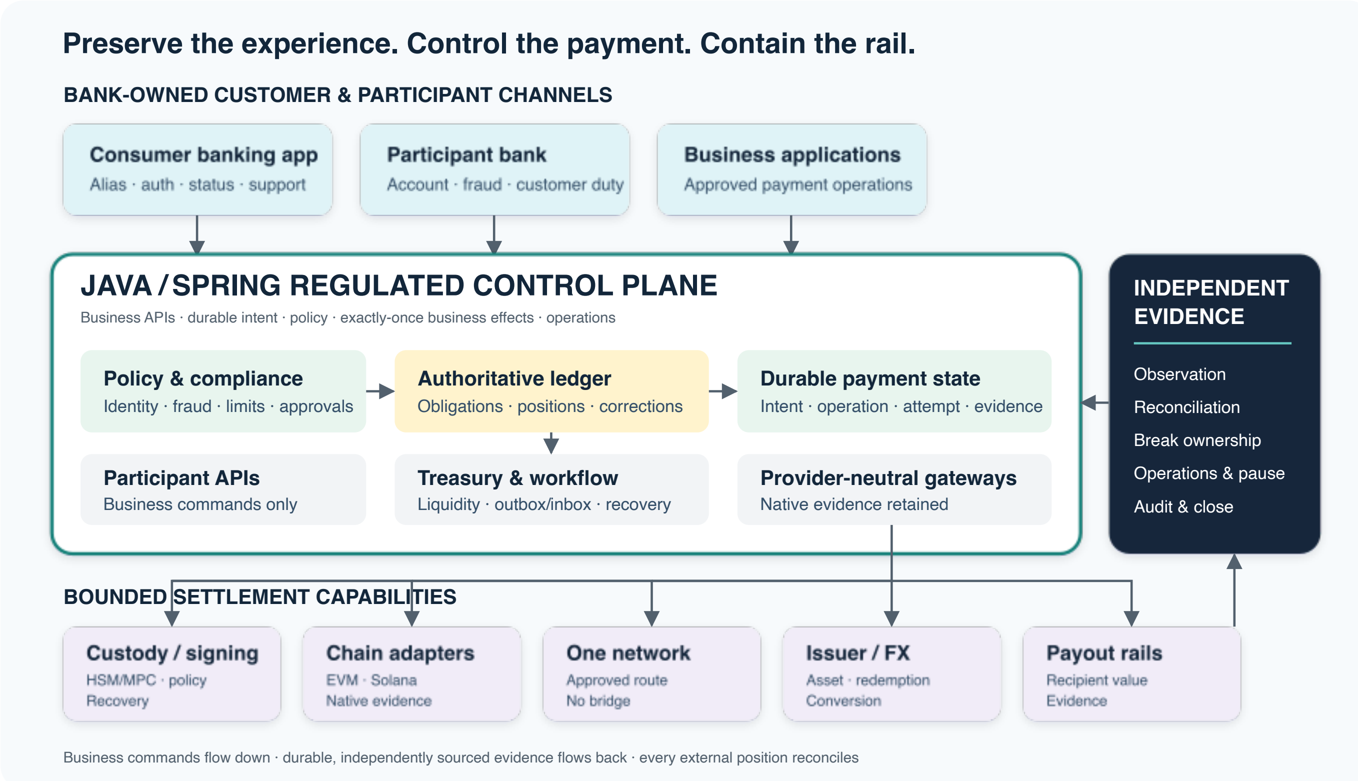

Figure 1: Executive ecosystem architecture. Bank-owned channels invoke business operations in a Java/Spring control plane. The internal ledger, durable state, policy, and reconciliation remain authoritative while custody, networks, issuers, FX, and payout rails remain bounded external systems.

3.3. Evidence, reconciliation, and operations

Reconciliation compares ledger and workflow records with chain data, custody balances, issuer statements, bank accounts, FX trades, fees, and payout outcomes. Every break has identity, owner, severity, age, evidence, disposition, and governed repair. Operations receives pause, replay, investigate, compensate, rotate, and exit capabilities at business boundaries rather than unrestricted database or RPC access.

> **The architecture is intentionally asymmetric.** The regulated platform owns business truth and recovery. External systems own only their bounded native facts. Every boundary preserves enough evidence to explain disagreement.

4. Ethereum vs Solana

Dual-chain implementation status—commit 173ebcb

REFERENCE ARCHITECTURE	One vendor-neutral business architecture remains authoritative. Ethereum and Solana are parallel implementation profiles, not competing definitions of the product.
IMPLEMENTED / TESTED / LOCAL DEMO	The same Spring control plane, PostgreSQL workflow persistence, local signing boundary, four-account reserve/liability ledger, reconciliation, and replay controls run through Web3j to Anvil or through Sava to a private Agave validator. The pinned baseline records 538 tests discovered, 536 executed, two opt-in native-validator tests skipped by default, and zero failures or errors.
PRODUCTION GATE	Real bank and reserve evidence, issuer and custody authority, HSM/MPC, public-network providers and observers, approved accounting, legal/compliance decisions, customer remedies, disaster recovery, and operating evidence remain outside the local implementation.

Every current implementation statement in this section is pinned to `johnwhitton/digital-banking@173ebcbb002cacad479c7ced4361106e7c6f21dc`. The environment is synthetic and loopback-only: it moves no real customer money, submits to no public network, and establishes no Early Warning Services or Zelle deployment claim.

The profiles implement **identical business semantics** through **intentionally different native blockchain mechanisms**. They use the **same Java control plane**, **different chain adapters**, the **same accounting model**, the **same reconciliation model**, and the **same replay semantics**. Native identity, signing, lifetime, instruction, and observation evidence deliberately remain chain-specific.

4.1. Three business flows, one control model

Business flow	Shared semantics	Native execution
Traditional bank transfer	Bank-owned transfer and settlement processing; shared identity, idempotency, ledger, notification, and reconciliation patterns.	No stablecoin or blockchain effect. The local reference implementation has no Zelle, ACH, RTP, or FedNow adapter.
USD to USDZELLE	Durable acquisition, synthetic funding evidence, policy and signing authority, mint observation, accounting, and reconciliation. HTTP acceptance is not mint completion.	Ethereum calls the role-gated Solidity contract through Web3j. Solana uses an associated token account and `MintToChecked` through Sava.
USDZELLE to USD	Durable custody, payout before burn, supply evidence, accounting, and reconciliation. Burn finality is not customer-visible USD completion.	Ethereum transfers to ADMIN custody and burns the caller-held balance. Solana uses `TransferChecked` to custody and `BurnChecked` through Sava.

4.2. Native mechanisms remain explicit

Concern	Ethereum profile	Solana profile
Runtime	Solidity `LocalReferenceToken`; Web3j adapter; local Anvil chain.	Classic SPL Token Program; Sava adapter; local Agave validator.
Signing	secp256k1 transaction signing under contract-role authority.	Ed25519 signatures for fee payer and mint authority or token-account owner.
Execution	Contract mint, transfer, and own-balance burn; nonce, gas, transaction hash, receipt, and event evidence.	Canonical associated token accounts; `MintToChecked`, `TransferChecked`, and `BurnChecked`; blockhash lifetime, signature, account, instruction, and slot evidence.
Observation	Receipt/event evidence is reconciled independently from submission.	A separate observer requires exact instructions and account/supply deltas at `finalized` commitment.
Recovery	Persisted intent and attempts prevent blind resubmission and support replay after restart.	The same rule applies; retained signatures are inquired before any safe, parented replacement after proven expiry.

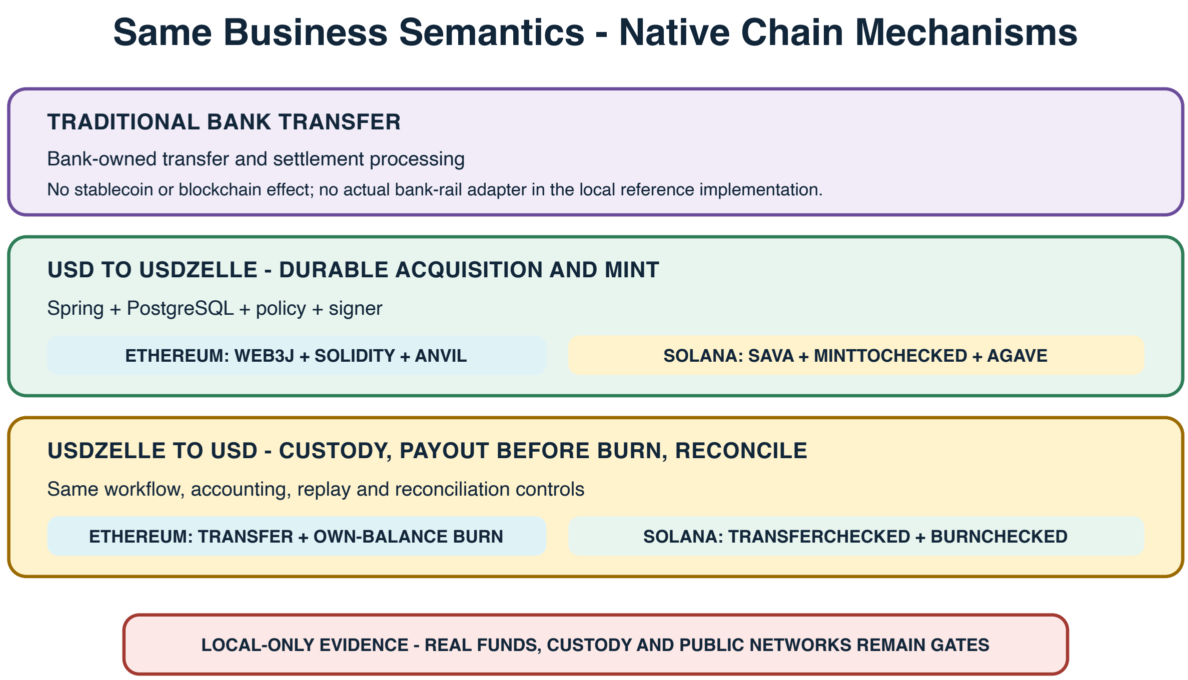

Figure 2: Three business flows keep one financial meaning while Ethereum and Solana use native execution mechanisms.

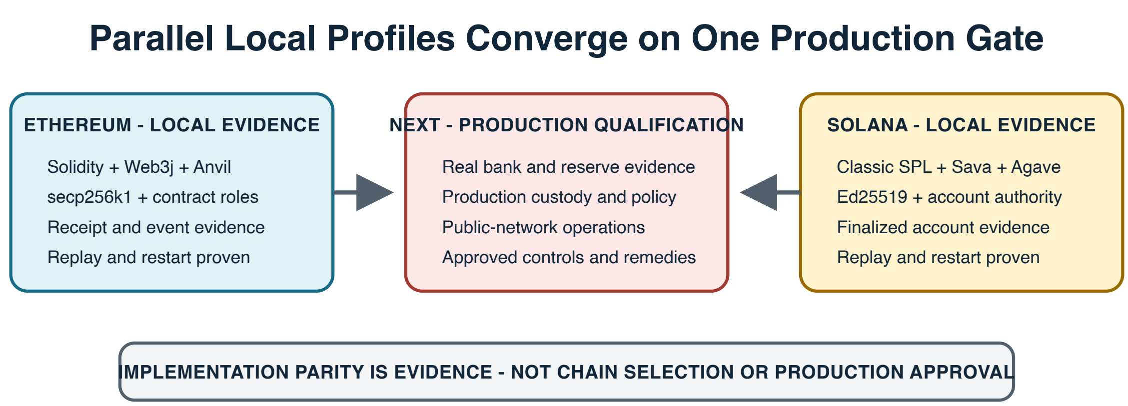

Figure 3: Both local implementation profiles are evidence-bearing; production qualification is the next independent gate.

Executive implication

Implementation parity reduces architectural uncertainty; it does not select a network or establish production readiness. The next work should qualify one bounded route with real authority, custody, bank/reserve, public-network, accounting, recovery, and operating evidence before expanding scope.

Detailed claim-to-source/test mappings are in the **Solana implementation traceability matrix** and **Solana implementation/publication alignment report**.

5. Ten decisions that keep the platform governable

These decisions are the executive control surface. Each recommendation points to a canonical ADR; the brief summarizes its business consequence without creating a second authority. Profile-specific items remain proposed until their named gate closes.

01 · The internal ledger remains authoritative PROPOSED

Recommendation. Record every obligation, reservation, fee, position, suspense item, and correction in a controlled double-entry ledger. Treat all external records as reconciliation evidence.
Why. Only the platform ledger spans the complete payment and its repair history. **Tradeoff.** The platform owns financial correctness, close, replay, and repair complexity.
Alternatives rejected. Chain state, a provider database, or a bank statement as platform truth. **Residual risk.** Posting defects and incomplete reconciliation.
Gate. Balance, precision, concurrency, replay, correction, restore, and close tests with financial review. **ADR-001**.

02 · Blockchain is an external settlement leg PROPOSED

Recommendation. Submit only approved business operations through chain adapters; retain native transaction and observation evidence.
Why. A chain can establish its own state, not customer entitlement, bank payout, accounting close, or legal discharge. **Tradeoff.** Adapters and reconciliation add engineering work.
Alternatives rejected. Chain as the payment system of record or an unrestricted privileged RPC gateway. **Residual risk.** Providers or observers can disagree.
Gate. Independent observation, ambiguous-outcome recovery, provider-failure drill, and exit test. **ADR-013**.

03 · Four finalities remain independent PROPOSED

Recommendation. Model customer-visible completion, accounting finality, blockchain finality, and legal settlement finality as independent judgments with named evidence and policy.
Why. The four authorities can advance or diverge independently. **Tradeoff.** APIs, storage, support tooling, and operations become more explicit.
Alternatives rejected. One `settled` flag or a simple business/chain split. **Residual risk.** Users may still misinterpret a status.
Gate. Test every pairwise divergence; obtain accounting and legal approval for route-specific evidence. **ADR-002**.

Evidence status: PROPOSED canonical architecture decision from ADR-002. Economic finality is a chain-risk threshold inside blockchain policy, not a substitute for legal settlement finality.

04 · Customer blockchain complexity remains invisible PROFILE DECISION

Recommendation. Preserve the bank-owned experience; do not require consumers to hold tokens, manage seed phrases, pay gas, or understand chain addresses in the initial pilot.
Why. The pilot should isolate settlement learning from a new custody, recovery, disclosure, and support product. **Tradeoff.** The platform retains backend responsibility.
Alternatives rejected. Consumer token or wallet at pilot launch. **Residual risk.** Invisible failures can still affect customers.
Gate. Journey, disclosure, support, dispute, and loss-allocation review. **ADR-003**; `SLOT-WALLET-001`.

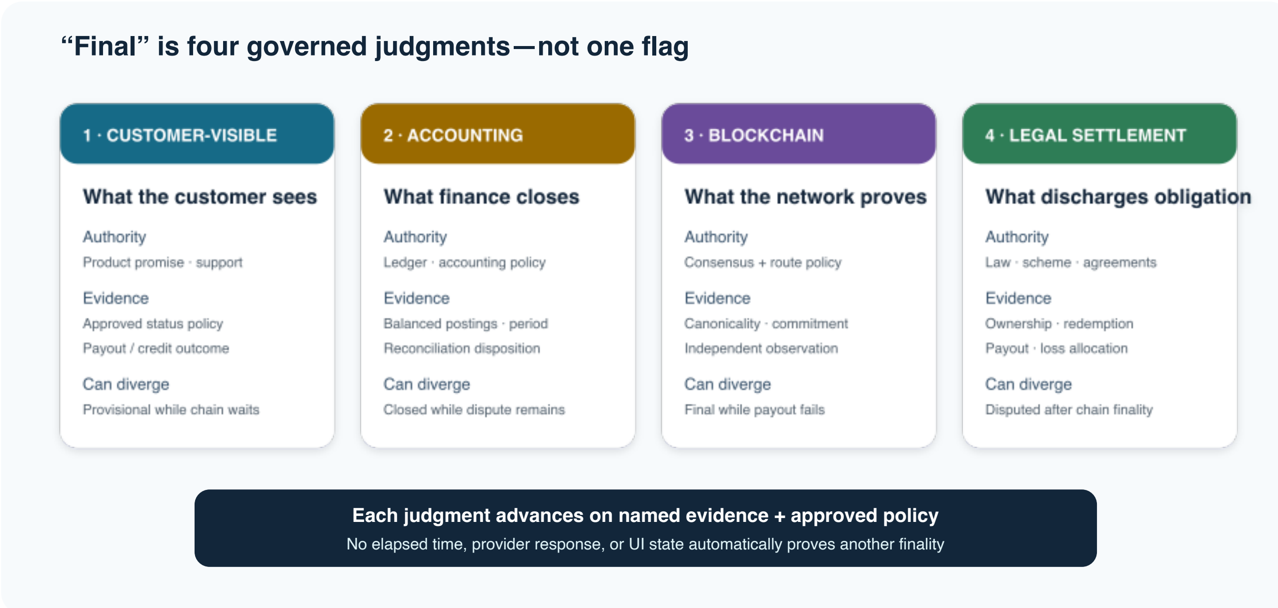

Figure 4: Four independent finalities. Each judgment has its own authority, evidence, customer or operational consequence, and correction path.

05 · Java/Spring owns enterprise orchestration

PROFILE DECISION

Recommendation. Use Java/Spring for participant APIs, durable intent, policy, ledger integration, messaging, reconciliation, observability, and operations.
Why. These are long-lived enterprise controls with transactional and operational demands. **Tradeoff.** The platform remains polyglot at native chain boundaries.
Alternatives rejected. Moving enterprise authority into chain-native services or forcing contracts/programs into Java.
Residual risk. Schema and adapter drift.
Gate. Contract tests, threat model, observability, load test, and failure drills. **ADR-004**; `SLOT-EXEC-001`.

06 · Chain technology stays behind adapters

PROPOSED

Recommendation. Keep EVM JSON-RPC, Solana commitment/blockhash rules, custody SDKs, and provider types behind business-level contracts.
Why. Intent, ledger, idempotency, and finality invariants should survive a chain or provider change. **Tradeoff.** A lowest-common-denominator abstraction would be unsafe. Native evidence must remain available.
Alternatives rejected. Vendor SDK types in the domain or an arbitrary contract-call API. **Residual risk.** Lossy abstraction and privileged escape paths.
Gate. Native dossier, policy allowlists, conformance tests, and replacement adapter. **ADR-005**.

07 · Start with one asset, one network, one corridor

PROFILE DECISION

Recommendation. Limit participants, notional, duration, asset, network, corridor, and operating windows; prefund liquidity and expand one dimension at a time.
Why. A narrow route produces interpretable evidence and a credible rollback. **Tradeoff.** Learning is intentionally incomplete and prefunding consumes capital.
Alternatives rejected. Multi-asset/network launch or a demo that never exercises participant operations. **Residual risk.** Results may not generalize.
Gate. Entry criteria, success measures, stop conditions, exit rehearsal, and independent approval. **ADR-006**; `SLOT-ROADMAP-001`.

08 · No bridge by default

PROFILE DECISION

Recommendation. Keep the initial route on one native network with no generic bridge.
Why. A bridge adds contracts or programs, validators or attesters, liquidity, replay, finality composition, governance, incident, and exit risk. **Tradeoff.** Reach and network flexibility are constrained.
Alternatives rejected. A generic cross-chain route in the pilot; issuer-native interoperability remains a later evaluation.
Residual risk. Single-network concentration.
Gate. Automated topology scan and a separate architecture/security/treasury/legal decision for any exception. **ADR-015**.

09 · Qualified custody or HSM/MPC controls signing

PROFILE DECISION

Recommendation. Bind signing to policy, transaction identity, allowlists, quorum, dual control, recovery, audit, and provider exit. Never place production keys in application memory.
Why. Cryptographic authority is a financial control, not an SDK convenience. **Tradeoff.** Integration, ceremony, and diligence cost increase.
Alternatives rejected. Application keys, direct consumer signing, or custody labels without control analysis. **Residual risk.** Authentication, recovery, and concentration can defeat nominal MPC.
Gate. Compromise, pause, rotation, recovery, export, and in-flight migration drills. **ADR-012**; `SLOT-CUSTODY-001`.

10 · L2 or issued stablecoin only after evidence

PROFILE DECISION

Recommendation. Do not build a proprietary L2 or issue an organization-specific stablecoin for the pilot. Reopen either choice only when measured need justifies the new obligations.
Why. An L2 introduces sequencer, settlement, data-availability, bridge, upgrade, gas, governance, and exit duties. Issuance introduces reserve, redemption, legal, compliance, and loss-allocation duties. **Tradeoff.** Near-term strategic control is lower.
Alternatives rejected. Build now for hypothetical scale or permanently prohibit future evaluation. **Residual risk.** Strategic pressure can bypass gates.
Gate. Separate business case and ADR with legal, security, treasury, operations, recovery, and exit proof. **ADR-010**; `SLOT-L2-001`.

6. Chain evaluation: choose the route, not the brand

VALIDATE No public evidence establishes a Zelle chain selection. The correct executive decision is therefore a gated comparison tied to the pilot asset, corridor, participants, legal rights, liquidity, custody support, privacy posture, failure tolerance, and exit plan. A universal chain winner would be less rigorous than a route-specific decision.

Evidence status: PUBLIC FACT narrow protocol facts refreshed 16 July 2026 (Ethereum Foundation 2026b; Base 2026a; Base 2026b; Solana Foundation 2026c; Solana Foundation 2026a); PROPOSED qualitative working hypotheses; PROFILE DECISION candidate disposition.

Table 1: Decision summary; the complete dimension record is in `data/chain-comparison.csv`.

Candidate	Cost posture	Finality posture	Engineering fit	Principal operating risk	Disposition
Ethereum	Variable L1 gas; benchmark	Native PoS justified/finalized states	Mature EVM/Solidity; direct JSON-RPC/Web3j	Fee/capacity variability and public governance	Eligible baseline; not selected
Base	Lower execution target plus L1 data/security fee	L2 inclusion then L1 publication/finality; withdrawals differ	EVM compatible; direct JSON-RPC/Web3j	Sequencer, L2 governance, and L1 dependency	Eligible candidate; not selected
Solana	Low base-fee design plus optional priority fee	Processed, confirmed, and finalized commitments	Rust/Anchor native; neutral service boundary for Java	Commitment/RPC/account contention and native skills	Eligible candidate; not selected
Future Zelle L2	Unknown until workload and stack design	Must compose sequencer, settlement, DA, and exit	Could be EVM-compatible; full platform must be built	Highest governance, bridge, security, and exit burden	Not applicable to pilot

6.1. What the public sources establish

PUBLIC FACT Ethereum documents transaction progression through inclusion, justification, and finalization. (Ethereum Foundation 2026b) “Finalized” remains a consensus judgment, not legal discharge.

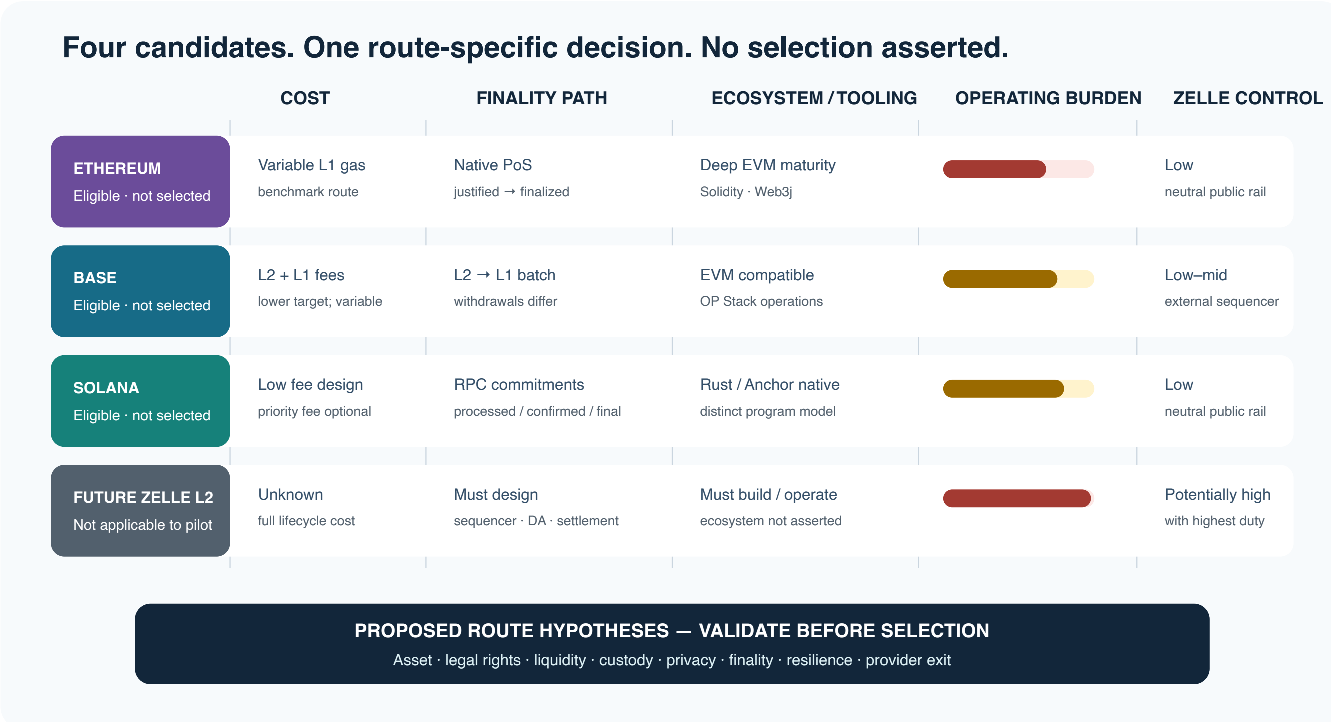

Figure 5: Executive chain comparison. Narrow protocol facts inform proposed qualitative working hypotheses across cost, finality, tooling, operating, and control dimensions. The bands are not verified comparative facts, measured scores, or a selection.

PUBLIC FACT Base documents a two-part fee model: L2 execution plus L1 security/data cost. Its finality documentation separates fast sequencer/L2 inclusion from L1 batch publication and batch finality, with distinct L2-to-L1 withdrawal behavior. (Base 2026a; Base 2026b)

PUBLIC FACT Solana exposes processed, confirmed, and finalized RPC commitments. Its fee model combines a base fee with an optional compute-unit priority fee. (Solana Foundation 2026c; Solana Foundation 2026a)

PROPOSED The remaining comparative judgments—including ecosystem maturity, throughput posture, custody support, operating burden, governance exposure, and strategic control—are decision hypotheses. They are included to expose diligence questions, not to establish current facts or a winner. Before selection they require authoritative refresh, route benchmarks, provider diligence, native-domain review, sensitivity analysis, and profile approval. Low nominal fees or headline throughput cannot replace failure drills and end-to-end route evidence.

PROFILE DECISION A future Zelle L2 is a decision slot, not a candidate for the minimum pilot. Strategic control must be weighed against the obligation to own or govern sequencing, settlement, data availability, upgrades, fees, infrastructure, monitoring, incident response, interoperability, and exit. “Control” alone is not a business case.

6.2. Promotion gates

Before selection, the decision team must demonstrate: the regulated asset is natively available and redeemable for the intended entities; legal and accounting treatment is approved; finality thresholds map to the four-finality model; peak and stress cost/latency are measured; custody and observation paths satisfy independence and recovery; privacy exposure is accepted; liquidity is available; and an alternate or termination path can close all obligations. The profile records the result under `SLOT-CHAIN-001`; the neutral architecture does not change.

7. Recommended pilot: a bounded route with a real exit

PROFILE DECISION The minimum credible pilot is one approved regulated stablecoin, one approved public network, one corridor, and a limited participant cohort. It should move enough controlled value to exercise treasury, signing, chain observation, payout, support, and accounting—but remain small enough to pause, reconcile, compensate, and terminate without systemic consequence.

Evidence status: **PROFILE DECISION** proposed Zelle-profile pilot. Asset, chain, corridor, participants, providers, and deployment remain validation gates.

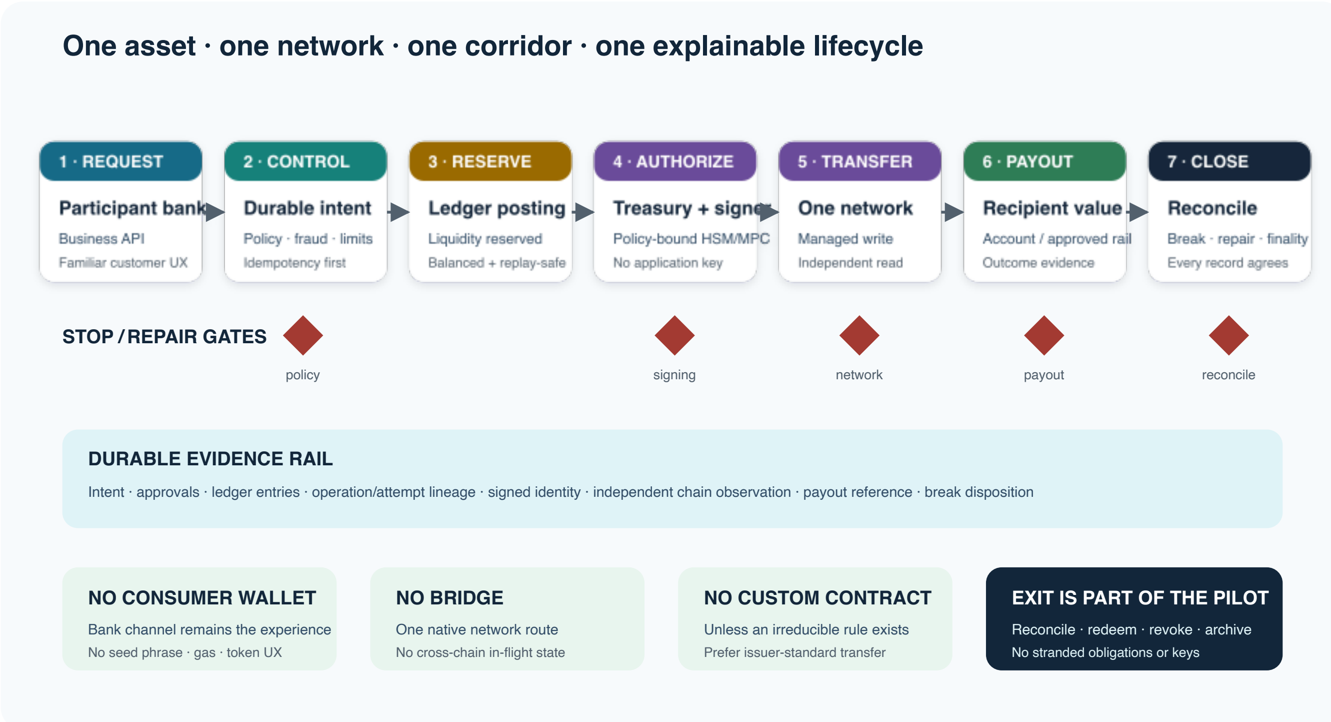

Figure 6: Bounded pilot flow. A durable intent and reserved ledger position precede compliance, signing, chain submission, independent observation, payout, and reconciliation. Every external step can pause or enter controlled repair.

7.1. Pilot envelope

Dimension	Proposed boundary
Asset	One regulated, redeemable stablecoin with approved issuer, reserve, freeze, redemption, and legal treatment. PUBLIC FACT USDC is an example of an issued multi-network asset, not a selection. (Circle 2026)
Network	One native public network selected through `SLOT-CHAIN-001`; no generic bridge and no cross-network in-flight obligation.
Corridor and cohort	One approved send/receive corridor; named participant banks or controlled internal entities; explicit customer, notional, velocity, and duration limits.
Liquidity	Prefunded treasury or omnibus positions as an operating assumption, with ownership, accounting, limits, rebalancing, and emergency exit approved before value moves.
Custody and infrastructure	Qualified custody or HSM/MPC; managed write submission paired with a materially independent observation path. No provider is preselected.
Customer boundary	No direct consumer wallet, token balance, gas, bridge, seed phrase, or chain-address workflow.
On-chain logic	Use issuer-standard transfer behavior; no custom contract or program unless an irreducible shared rule is demonstrated and separately reviewed.

7.2. Entry criteria

The pilot starts only after the legal entity and licensing map, asset rights, accounting treatment, participant agreements, customer treatment, sanctions/fraud model, custody authority, network policy, liquidity model, support playbook, and loss-allocation matrix are approved. The control plane must pass deterministic ledger, idempotency, ambiguity, reorg/finality, signing, reconciliation, restore, and pause tests. Production credentials and limits must be created through dual-control ceremonies.

7.3. Success, stop, and exit

Success is an evidence set, not a launch date: zero unbalanced postings or duplicate business effects; every intent and external attempt explainable; reconciliation breaks within approved aging; observed cost and latency within route thresholds; no policy bypass; recovery and provider failover demonstrated; support and participant outcomes accepted; and every finality judgment traceable to evidence and policy.

Stop new intents automatically or by governed operator action when legal approval changes, reserve/redemption confidence falls, the asset deviates beyond limits, liquidity buffers breach, signing authority is suspect, independent observers materially disagree, the network or provider exceeds outage/finality thresholds, reconciliation breaks age beyond tolerance, sanctions/fraud controls degrade, or customer loss allocation becomes unclear. Stopping new work does not abandon in-flight obligations: the state machine moves them to controlled completion, return, compensation, or manual repair.

Exit means more than switching off an API. The team must export addresses, balances, attempts, policies, approvals, and audit evidence; reconcile ledger, custody, issuer, chain, bank, and payout positions to zero or an approved retained balance; redeem or transfer the asset; revoke and archive signing authority; complete customer/participant treatment; and preserve records. Only a rehearsed exit makes a pilot genuinely bounded.

8. Wallet and custody recommendation: design roles before providers

"Wallet" is too broad to be an architecture decision. The platform must first name who can initiate, approve, sign, recover, pause, and exit for each value role. Only then can it evaluate custody technology or providers.

Table 2: Wallet and authority roles for the bounded pilot.

Role	Authority and experience	Pilot posture
Consumer	Initiates through the bank experience; no direct token/key authority	No consumer wallet
Participant bank	Authenticates customer, applies bank controls, invokes business APIs	No raw signer or arbitrary RPC
Settlement/operator	Holds narrowly scoped rail authority for approved intents	Policy-bound institutional wallet
Issuer/mint	Issues, redeems, freezes, or administers the asset under issuer rules	External authority; reconcile and monitor
Treasury	Funds, rebalances, pays fees, and manages exposure/limits	Separate duties, limits, and addresses
Administration	Configures allowlists, contracts/programs, policies, pause, and upgrades	Stronger quorum and change ceremony
Recovery	Restores or migrates authority under exceptional conditions	Offline/independent ceremony and tested exit

PROFILE DECISION The minimum pilot should not require direct user wallets. A custodial versus non-custodial consumer-wallet choice would add authentication, device binding, transaction consent, recovery, account portability, privacy, sanctions, support, and legal-custody questions without improving the initial settlement-learning objective. If a future product demonstrates customer value, it should reopen `SLOT-WALLET-001` as a separate product and architecture decision.

PROFILE DECISION Institutional settlement authority should use qualified custody or an HSM/MPC boundary with explicit policy. "MPC," "non-custodial," or "embedded" is not an adequate control description. The review must identify who authenticates, who holds shares or keys, who defines policy, whether a provider can act alone, how destination and amount allowlists bind to the signed payload, how recovery changes authority, and how the organization exits with complete evidence.

The signing request should carry the payment intent, operation and attempt identity, approved asset/network, contract or program, method/instruction, amount, destination, fee bounds, nonce or blockhash, expiry, policy version, and approvals. The signer returns a signature or transaction identity plus durable audit evidence. A lost response remains ambiguous until activity logs and network observation determine whether the original request was accepted.

Custody control test

Assume the application credential is stolen, one operator is malicious, the provider is unavailable, a recovery administrator is compromised, and the organization must migrate providers during in-flight payments. The design passes only if policy still prevents unauthorized value, legitimate obligations can be recovered, and evidence remains complete.

See **ADR-011** and **ADR-012**. Provider evaluation belongs in the reusable engineering reference library and a dated procurement/control review, not in the neutral architecture.

9. Java and the engineering boundary

PROFILE DECISION Java/Spring should own the regulated enterprise control plane: participant APIs, durable payment intent, authorization and policy coordination, ledger integration, transactional messaging, workflow, reconciliation, case/operations interfaces, telemetry, and controlled repair. This is a profile recommendation aligned with the architecture's responsibilities, not a claim about EWS's current stack.

PUBLIC FACT Spring's official documentation covers web, messaging, transactional data/persistence, integration, scheduling, caching, observability, and testing concerns relevant to long-lived enterprise services. (Spring Team 2026) The architecture benefit is not the brand itself. It is the ability to keep business authority, transactional boundaries, policy, and operational evidence in a mature control-plane environment with clear ownership.

Table 3: Use the language that owns each execution environment.

Boundary	Primary role	Why it belongs there
Java/Spring	Business API, intent, ledger, policy, workflow, messaging, reconciliation, operations	Owns regulated enterprise state and long-lived service contracts
Web3j / EVM client	Typed JSON-RPC, transaction, receipt, event, and contract-wrapper integration	Adapter library inside the Java boundary; not a chain or consensus engine
Solidity	Minimal EVM smart-contract rules when shared on-chain enforcement is irreducible	Native EVM execution and security semantics
Rust / Anchor	Solana programs and native services where the Solana model requires them	Native account, instruction, CPI, compute, and deployment semantics
TypeScript or Rust service	Optional native adapter when ecosystem support is materially stronger	Remains behind the same language-neutral domain contract

PUBLIC FACT Web3j describes itself as a Java library for Ethereum JSON-RPC and smart-contract integration. (Web3j 2026) It can encode transactions, call nodes, consume receipts/events, and generate typed wrappers. It does not implement Ethereum's public consensus, establish legal finality, or make Java an EVM contract language. Production signing should be delegated to the approved custody/HSM boundary rather than implemented with application-held credentials.

Solidity remains the native language for EVM contracts. (Solidity Team 2026) The pilot should avoid a custom contract unless the issuer-standard transfer model cannot express an irreducible shared rule. If a contract is required, its scope, upgrade/admin authority, audit, monitoring, pause, and exit are separate security decisions.

PUBLIC FACT Solana documents Rust as the primary language for programs compiled to its chain-native sBPF environment. (Solana Foundation 2026b) A serious Solana option therefore requires first-class Rust/Anchor, account/PDA, instruction/CPI, token, compute-budget, transaction-lifetime, RPC commitment, testing, deployment, and monitoring expertise. An all-Java mandate would either ignore native semantics or force an immature abstraction. The control plane can remain Java while a neutral adapter contract permits a native implementation.

Boundary rule. Java owns the payment; native technology owns the chain interaction; the adapter contract ensures neither can silently take authority from the other.

10. Roadmap: expand evidence before scope

The roadmap treats every phase as an investment option with explicit promotion and stop gates. It does not commit to production scale before the preceding phase proves financial correctness, authority, recoverability, and an end-to-end business outcome.

The local Ethereum and Solana profiles now supply parallel implementation evidence for the control-plane, persistence, signing, chain-adapter, observation, replay, accounting, and reconciliation boundaries. They do not select a production chain. The roadmap therefore advances from mechanism proof to qualification of one bounded route under real financial and operational controls.

Evidence status: PROFILE DECISION proposed implementation roadmap under `SLOT-ROADMAP-001`; dates and organization commitments remain unknown.

1. **Discovery and control definition.** Select candidate corridor and participants; map legal entities, obligations, systems of record, customer treatment, data, risks, and authority boundaries. Exit if the business benefit or legal path is not credible.

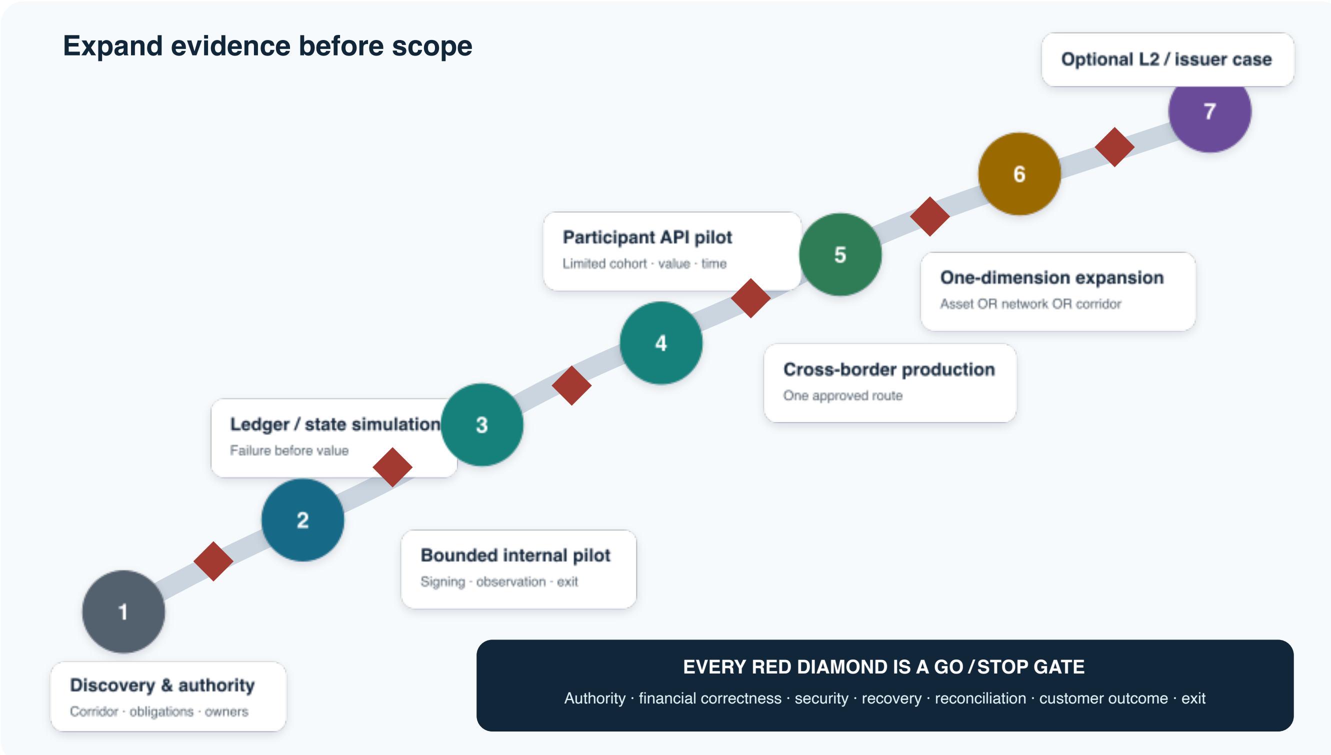

Figure 7: Evidence-gated roadmap. Discovery and simulation precede bounded value; participant production precedes multi-asset or multi-network expansion; an L2 or issuer decision is last and optional.

2. **Ledger, state, and simulation.** Implement the exact-money ledger contract, durable intent, attempt lineage, outbox/inbox, four finalities, and reconciliation using synthetic data. Simulate duplicates, reordering, timeouts, reorgs, provider disagreement, payout failure, corrections, and restore.
3. **Bounded internal pilot.** Integrate sandbox or controlled external systems, institutional signing, managed submission, independent reads, treasury, and operations. Exercise the complete stop and exit process before customer or participant exposure.
4. **Participant API pilot.** Add a limited participant cohort behind business APIs, contractual limits, customer/support treatment, production controls, and high-touch monitoring. Expand only after success measures remain stable through normal and adverse conditions.
5. **Cross-border production.** Promote one corridor with approved service levels, capacity, on-call ownership, reconciliation close, compliance/fraud governance, financial controls, and continuity. Retain explicit route termination and provider exit.
6. **Multi-asset or network expansion.** Add one dimension at a time. Each asset, network, corridor, participant class, custody model, or provider receives an impact review and route-specific evidence. Cross-network movement is a new architecture decision.
7. **Evidence-based L2 or issuer decision.** Only measured limitations can justify a proprietary L2 or organization-issued stablecoin. Evaluate the full lifecycle, not just transaction cost or control. Deferral or rejection is a successful outcome when evidence does not justify the obligation.

Portfolio rule

Never expand asset, network, corridor, participant population, custody authority, and notional limit in the same release. Changing one dimension preserves the ability to attribute outcomes and roll back safely.

11. Principal risks and executive control posture

Risk is not a separate appendix to the architecture. It determines the authority model, state machine, pilot limits, provider boundaries, operating evidence, and stop conditions. The register below summarizes the minimum executive posture; `data/risks.csv` preserves stable IDs, owners, residual risk, and validation gates.

Table 4. Principal risk ownership, controls, and residual exposure.

ID	Risk	Control posture	Residual / gate
001	Regulatory and licensing	Map entities, activities, corridor, asset, counterparties, disclosures, and legal finality before design approval.	High until qualified written disposition; no value before approval.
002	Issuer and reserve	Diligence reserve/redemption/freeze rights; concentration limits; redemption and alternate-exit tests.	Issuer dependency remains; depeg/freeze scenarios must pass.
003	Chain availability and finality	Network-specific policy, independent reads, queuing, pause, reorg and governance playbooks.	Public-network risk remains; chaos and contradiction drills must pass.
004	Custody and key authority	HSM/MPC or qualified custody, dual control, payload policy, allowlists, ceremonies, recovery, and exit.	Concentrated authority remains; compromise/rotation/export rehearsals must pass.
005	Sanctions and fraud	Identity, screening, transaction monitoring, velocity/amount controls, holds, and case management.	Adaptive adversaries and false positives remain; model and case drills required.
006	Liquidity	Forecasts, prefunding limits, buffers, alerts, rebalancing, fees, and governed stop thresholds.	Capital cost and stress risk remain; emergency rebalance must pass.
007	Privacy	Minimize on-chain/provider data; address strategy, access, retention, and correlation-risk review.	Public metadata cannot be eliminated; data-flow approval required.
008	Provider concentration	Neutral contracts, independent observation, evidence export, replacement integration, and termination assistance.	Specialist dependencies remain; failover and migration must pass.
009	Smart-contract	Prefer issuer-standard transfer; minimize custom logic; verify code/admin authority; audit, allowlist, monitor, pause.	Token contract risk remains; exact deployed artifact review required.
010	Bridge	No bridge in the pilot; separate threat model and ADR for any exception.	Low in pilot only if topology confirms absence.
011	Reconciliation	Independent records, first-class breaks, aging/ownership, immutable correction, daily close, and restore.	Late/conflicting data remains; repair drills and aging SLO required.
012	Customer loss allocation	Explicit status, agreements, support, disputes, reversal/compensation rules, and scenario matrix.	High until every partial failure has approved treatment.

PUBLIC FACT FATF's current stablecoin work highlights illicit-finance risks involving stablecoins, unhosted wallets, and cross-chain activity. (Financial Action Task Force 2026) The response is not to assume that blockchain analytics substitutes for regulated controls. Identity, sanctions, fraud, transaction context, issuer authority, wallet control, chain evidence, payout, and customer treatment must work as an end-to-end decision system with reviewable model governance.

11.1. Failure containment priorities

Prevent duplicate value. Durable idempotency, uniqueness, attempt lineage, outbox/inbox, and reconciliation enforce one permitted economic effect even when delivery repeats. A timeout after possible submission is not permission to create a fresh transfer.

Separate pause from repair. A pause blocks new intents or a bounded operation class. It does not mutate history or erase obligations. In-flight work moves to observation, controlled completion, return, compensation, or manual repair with dual control.

Preserve raw evidence. Provider-normalized success is insufficient for audit and recovery. Retain request identity, signed payload or digest, network transaction identity, observation source and policy, block/slot context, issuer/custody records, payout references, and operator decisions.

Design exit before adoption. Provider or chain exit requires portable addresses, policy and audit history, replacement signing/submission/observation, in-flight handling, asset redemption or transfer, and a ledger-to-external close. A nominally non-custodial service can still be operationally non-portable.

Executive stop principle

When authority, evidence, or loss allocation is unclear, stop creating new obligations. Availability targets never authorize the platform to guess whether value moved.

12. First 90 days: Principal Engineer plan

PROFILE DECISION The first ninety days should reduce irreversible uncertainty before moving value. The deliverable is a jointly owned set of authority maps, decision records, executable contracts, simulations, and pilot gates—not a premature chain integration.

Table 5: First-90-day outcomes.

Window	Focus	Evidence produced
Days 1–30	Authority and current-state discovery	System-of-record map; participant/customer journey; legal-entity and obligation map; data and trust boundaries; current controls; decision and unknown register; candidate corridor and participant criteria
Days 31–60	Contracts and simulation	Exact-money ledger model; durable intent/state/finality contracts; API and event schemas; signing policy; reconciliation model; synthetic happy and failure paths; chain/custody evaluation scorecards
Days 61–90	Pilot readiness and operating proof	Candidate route dossier; measured sandbox behavior; ambiguity/re-org/provider/custody/restore drills; support and loss-allocation scenarios; limits and stop conditions; phased plan and go/no-go review

12.1. 1. Validate architecture and authority boundaries

Interview product, participant, payment operations, fraud, compliance, legal, treasury, finance, security, SRE, support, and platform owners. Trace one representative payment and every exception across systems. Name who can authorize, observe, reverse, compensate, pause, recover, and declare each form of finality. Convert conflicts into explicit decisions rather than smoothing them into a diagram.

12.2. 2. Define systems of record and canonical decisions

Confirm the ledger authority and chart-of-account implications; identify current customer, participant, fraud, case, treasury, and settlement records; and establish a single ADR for each material choice. Revalidate the profile rather than assuming Phase 0 recommendations are organization decisions. Record chain, asset, custody, infrastructure, deployment, corridor, and participant selections as open gates.

12.3. 3. Build financial and failure simulation first

Implement executable ledger postings and state transitions with synthetic data before any production signing. Simulate duplicate requests, redelivery, out-of-order events, lost responses, partial approval, network congestion, reorg or commitment disagreement, issuer freeze, signer outage, payout failure after chain finality, liquidity depletion, period close, correction, backup restore, and provider exit. Every scenario must preserve balance, lineage, evidence, and a governed next action.

12.4. 4. Select the pilot corridor and controls

Score candidate corridors on customer value, participant readiness, legal clarity, currency and payout access, stablecoin rights, liquidity, fraud/sanctions exposure, privacy, operational support, and ability to exit. Then evaluate asset, chain, custody, and providers for that route. The ordering matters: selecting a chain before the obligation and corridor would optimize the wrong system.

Day-90 decision

Proceed only if the route has a credible business benefit, approved authority, balanced and replay-safe accounting, explicit four-finality policy, controlled signing, independently observable network evidence, complete reconciliation, rehearsed stop/exit, and named owners. Otherwise continue discovery or stop without treating schedule as failure.

13. Relationship to the engineering series

This brief is the first organization-profile edition of a vendor-neutral publication contract. It exports a decision narrative, pilot posture, risk frame, and first-90-day roadmap for an executive audience. It does not become the authority for ledger semantics, finality, detailed chain engineering, or implementation.

Series artifact	Relationship	Authority
Normative kernel	Owns evidence taxonomy, invariants, publication/profile contracts, governance, and release gates	Level 0
Volume I: Reference Architecture	Canonical layered architecture, ledger/state/finality model, flows, security, reconciliation, technology boundary, pilot, and roadmap	Architecture
Canonical ADR library	Owns each decision once; this brief links rather than duplicates authority	Decision
PROFILE-ZELLE	Applies registered chain, wallet, custody, SDK, infrastructure, roadmap, L2, and executive-tailoring slots	Configuration
Volume II: Engineering Companion	Owns detailed engineering patterns, native chain dossiers, implementation guidance, tests, deployment, and runbooks	Engineering
Volume III: Reference Implementation	Maps the neutral architecture to runnable unbranded code and explicit production gaps	Implementation

The deeper path for a reviewer is deliberate: start here for the recommendation; use Volume I to challenge the architecture; use the ADRs to examine alternatives and consequences; use the Zelle profile to see which values are organization-specific; use Volume II for engineering patterns; and use Volume III for code mapping and production gaps. The claim ledger in this release preserves the exact public, inferred, proposed, profile-specific, assumed, and unknown boundaries behind the executive wording.

What I would build first if hired. Not a token and not a bridge: an executable ledger and payment-intent model, a failure simulator, and reconciliation that can prove what happened. Those controls make every later chain, custody, and corridor choice safer and faster.

Release scope. This Ethereum-aligned patch updates the Executive Brief and linked Volume I profile without changing Volume II or Volume III. All organization choices remain subject to qualified internal review and the registered profile gates.

A. Transaction-cost decision appendix

This appendix turns headline chain fees into a decision gate. It is a concise consumer of the series Engineering Reference Library module *Chain Transaction-Cost and Operating-Economics Analysis* (`KB-CHAIN-002`), version 1.1.1 as of 2026-07-18. It does not select a chain, approve an L2, forecast fees, or replace a pilot benchmark and commercial quotes.

Executive conclusion. There is no universal cost winner. Compare the cost per completed, reconciled business effect at the required finality—not the cheapest receipt under one observed block.

A.1. Comparable reference case

PUBLIC FACT (Ethereum Foundation 2026a; Base 2026a; Solana Foundation 2026a) Ethereum charges gas used at base plus priority fee; Base adds a separate L1 security/data component; Solana separates a signature fee from an optional CU-limit-based priority fee. The table applies those formulas to explicit planning assumptions. Contract gas, CU use/limit, serialized bytes, prices, retries, provider use, staffing, and low/reference/high bands remain assumptions or benchmark requirements.

Table 6: Modeled stablecoin-transfer network fee, USD.

Candidate	Low	Reference	High	Decision caveat
Ethereum	$0.0117	$0.1510	$8.3200	L1 fee exposure; mature EVM route still needs contract and custody measurement.
Base	$0.0008	$0.0071	$0.3960	L1 data exceeds modeled L2 execution in the reference case.
Solana	$0.0002	$0.0004	$0.0013	Low fee does not supply native-program, custody, RPC, or operating capability.
Future OP Stack L2	$0.0002	$0.0032	$0.1302	Hypothetical execution/data profile; fixed chain ownership is separate.
Future Orbit L2	$0.0002	$0.0032	$0.1302	Equal placeholder preserves uncertainty; it is not an equivalence claim.

The modeled Solana account-creation case separately requires about $0.153 of recoverable rent reserve at the reference price. An EVM ten-recipient batch can reduce fee per recipient, but it couples failure, idempotency, reconciliation, and compensation across the batch. Neither balance is visible in a simple “average fee” ranking.

A.2. Batching changes both unit cost and risk

Recommendation. Benchmark bounded application batching in the pilot, but keep every entitlement individually authorized, stateful, reconcilable, and repairable. Treat net settlement as a separate legal and accounting decision.

PUBLIC FACT (Agusx1211 et al. 2024; Solana Foundation 2026d) EVM supports batch-executor patterns and Solana supports atomic multi-instruction transactions, but neither source establishes a safe Zelle batch size, measured savings, custody fit, or production limit. The reproducible planning model evaluates sizes 1, 2, 5, 10, 25, 50, and 100. It stops Solana after a conservative ten-item modeled limit rather than extrapolating. EVM values through 100 remain hypotheses pending exact gas, calldata, bytecode, provider, custody, latency, value, and failure tests.

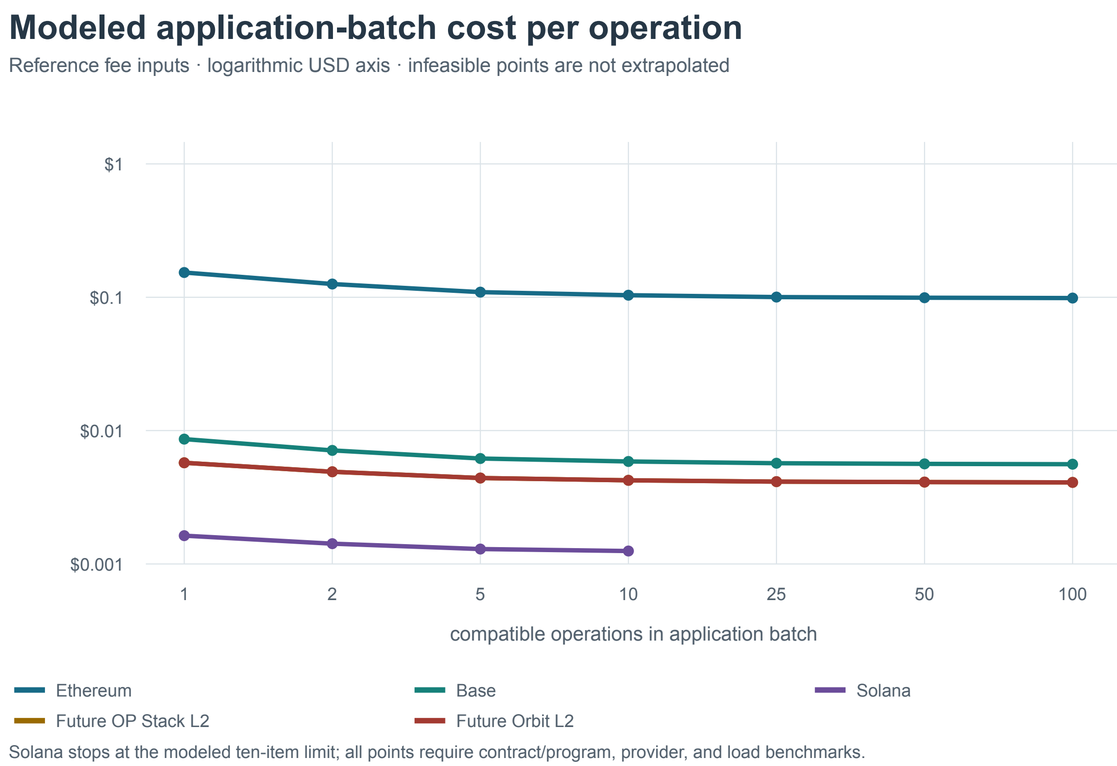

Figure 8: Modeled application-batch cost per operation. Curves include provider allowance, expected fee retry, and allocated first-use cost. They show shape, not production savings or a chain ranking.

Evidence status: PROPOSED transparent planning assumptions; VALIDATE route benchmark, burst behavior, provider and custody terms, and approved limits.

For 100 compatible operations every five seconds, a 10/1-second policy fills in 0.5 seconds, averages 0.25 seconds of added wait, and submits two batches per second. A 100/5-second policy fills at five seconds, averages 2.5 seconds of added wait and 50 queued operations, and submits 0.2 batches per second. The larger policy can also couple 100 operations to one full-batch failure. A production policy must flush earlier on value, risk, expiry, gas/CU/bytes, account or nonce conflict, maintenance, or operational limits.

What remains separate

One source transaction containing N cross-chain dispatches still creates N messages and destination effects. One future cross-chain message containing N destination calls changes caller, custody, approval, and failure semantics. Relayer delivery batching does not prove sender savings. The initial pilot adds none of those cross-chain dependencies; it measures chain-native application batching only.

The ledger retains each entitlement's intent, authorization, amount, lifecycle, hold, evidence, four-finality status, and accounting disposition. A batch adds a batch ID, policy version, item mapping, transaction/message/result evidence, gross/net totals, residuals, reconciliation, and repair or reversal records. Net settlement proceeds only after legal set-off or novation, liquidity, default, loss-allocation, customer timing, accounting, audit, and unwind rules are approved. A fee curve cannot approve those obligations.

A.3. Corrected decision economics

The prior $1.30, $1.16, and $1.15 figures at 100 thousand monthly transactions were too easy to misread. They are **fully allocated modeled operator costs, not gas fees**: a small pilot volume absorbs modeled platform, operations, provider, retry, and network costs. The architecture is unchanged; the economic label is corrected.

PUBLIC FACT (BaseScan 2026; Base 2026a) One observed 60-USDC Base transfer used 62,147 gas and paid 623,698,777,861 wei of L2 fee plus 2,266,222,878 wei of L1 fee. The total was 625,965,000,739 wei, or 0.000000625965000739 ETH—approximately $0.00116 at the explorer's historical conversion. This is one marginal network-fee observation, not a universal Base price, all-in payment cost, customer price, or forecast.

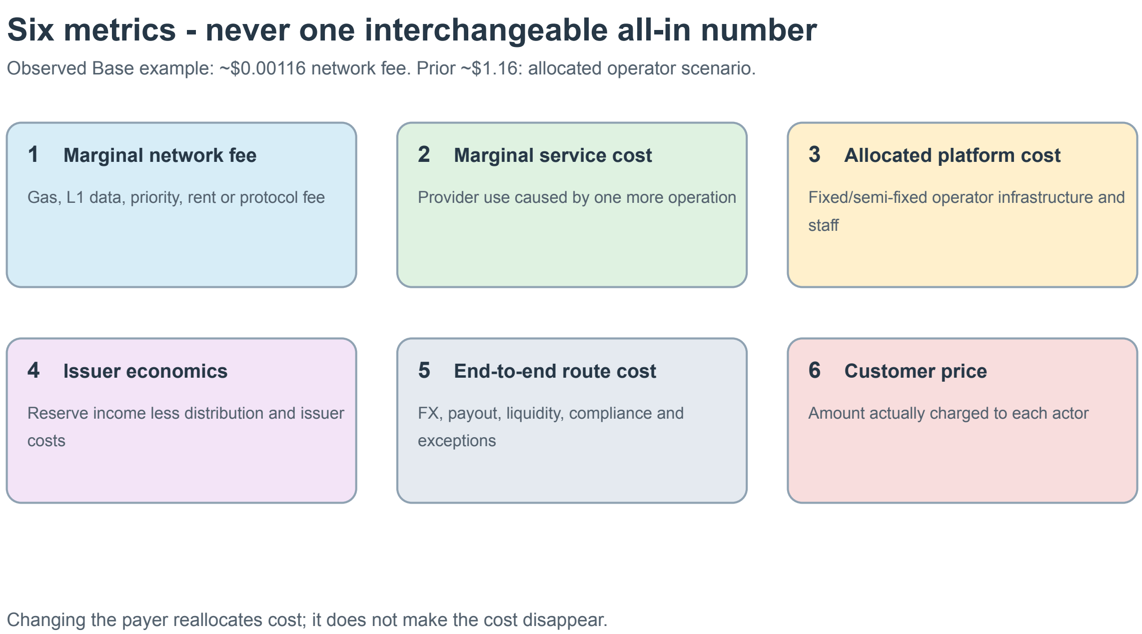

Figure 9: Six different economic questions. A network receipt, provider bill, platform allocation, issuer P&L, route cost, and customer price must never be presented as one interchangeable "transaction cost."

Evidence status: PUBLIC FACT (BaseScan 2026; Base 2026a) observed Base receipt and fee model; PROPOSED explicit metric boundaries; VALIDATE approved issuer, route, payer, provider contracts and customer price.

The transaction sender or sponsor initially pays the chain fee. Circle does not pay gas for every third-party USDC transfer. Zelle, a bank, wallet, exchange, merchant, issuer, or customer could fund or reimburse a route under an approved product and contract; changing the payer does not eliminate the economic cost.

Table 7: Business structures and operator economics.

Structure	Operator can recognize	Executive gate
External stablecoin	Approved platform, route, service, or contracted distributor revenue—not issuer reserve income by default	Issuer, asset rights, redemption, concentration, chain, contract, gas payer and exit diligence.
Operator-issued	Issuer economics in the correct entity plus approved payment-service economics	Entity, licensure, reserves, liquidity, custody, controls, distribution, tax, accounting and governance.
Co-issued / white-label	Contracted service and distributor share only; never both gross issuer income and the share	Issuer-of-record, principal/agent, brand, redemption, token controls, incidents, termination and migration.

For the immediate Zelle decision, an external asset remains the lowest-issuer- scope pilot structure; operator-issued and co-issued paths require explicit legal, treasury, risk, accounting and governance approval. This is a scope recommendation, not a finding about the unpublished ZelleUSD design.

Volume amortizes platform cost

At 100 thousand reference transactions per month, the fully allocated model produces about $1.30 per transaction for Ethereum, $1.16 Base, $1.15 Solana, and $3.26 for either future-L2 placeholder. The correct label is **Illustrative fully allocated operator-cost scenario—not a network-fee comparison**. At 100 million, the same outputs are approximately $0.151, $0.0092, $0.0025, and $0.0084. The result is not that one candidate "wins." It is that modeled fixed platform and operating capacity dominates a pilot, while marginal network cost matters more at scale and under stress.

Base and Solana's low marginal network fees were obscured by the pilot-volume fixed-cost allocation. RPC, custody, compliance, support, and staffing belong only to the actor operating or contracting for those functions; they are not protocol fees and are not automatically Zelle/EWS costs.

The all-in formula separates on-chain marginal fee, provider-variable use, failed/retried fee, fixed platform, modeled operations, liquidity/capital, and custody/signing. One-time implementation, audit, legal, migration, and launch remain separate investment costs.

Reserve income is not reserve principal

Reserve principal is not revenue; it backs the redemption liability. For $100 million average reserves, gross income is $2.0 million at 2 percent, $3.0 million at 3 percent, $3.5 million at 3.5 percent, $4.0 million at 4 percent, and $5.0 million at 5 percent before distribution, reserve management, custody, attestation, compliance, liquidity, staff, technology, insurance, tax, and other issuer costs.

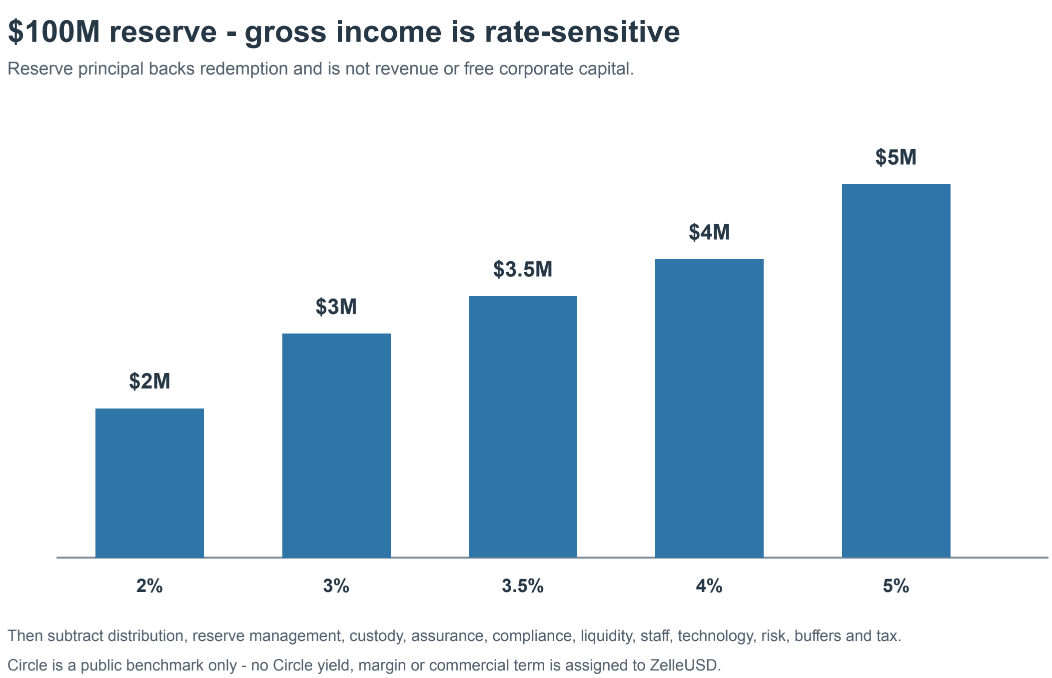

Figure 10: Illustrative $100 million reserve sensitivity. Gross yield is not issuer margin; the reference expense case remains negative across the shown range. This is not a ZelleUSD forecast or Circle margin.

Evidence status: **PROPOSED** explicit reserve and expense sensitivities; **PUBLIC FACT** (Circle Internet Group, Inc. 2026b; Circle Internet Group, Inc. 2026a; Circle Internet Group, Inc. 2026c) current Circle benchmark; **VALIDATE** approved issuer, reserves, realized yield, distribution, cost, tax and regulation.

PUBLIC FACT (Circle Internet Group, Inc. 2026b; Circle Internet Group, Inc. 2026a; Circle Internet Group, Inc. 2026c) Circle's Q1 2026 Form 10-Q reported $75.2 billion average USDC in circulation, a 3.5-percent reserve return, $653 million reserve income, and $405 million distribution and transaction costs. Reserve income was about 94 percent of reported total revenue and reserve income; a 66-basis-point year-over-year yield decline reduced quarterly reserve income by $106.3 million. This current official benchmark shows that reserve income is rate-sensitive and distribution cost can be material. Circle's scale, agreements, entities, products and accounting do not forecast ZelleUSD economics.

Sub-cent network fees may be economically minor beside reserve yield or the full payout, FX, liquidity, compliance, and exception route. Conversely, reserve yield falls when rates or average circulation fall, is not guaranteed, and can be materially consumed by distribution sharing and issuer costs.

What Zelle must decide before a business case

Choose the issuer and redemption obligor; external, operator-issued, or co-issued structure; payer and revenue recipient for every service; gas sponsor; RPC/custody/payout/FX sourcing; reserve and liquidity accounts; customer price; principal-versus-agent accounting; and legal, risk, treasury, tax, security and governance approvals. The model evaluates 2,160 structure, volume, payer, batching, chain, interest and operating-cost combinations, but it does not make those decisions.

A.4. What an owned L2 would actually commit Zelle to

PUBLIC FACT (Optimism Foundation 2026; Offchain Labs 2026) OP Stack fee mechanics separate execution, L1 data, and operator components; Nitro likewise separates L2 execution from parent-chain data and settlement concerns. A credible L2 case must price sequencer/batcher/proposer/challenger or validator infrastructure, L1 data and settlement, nodes and disaster recovery, RPC/indexer/ explorer, bridge and withdrawal, liquidity, upgrades, security, governance, incidents, staffing, vendor management, and exit.

PUBLIC FACT (Conduit 2026) Conduit's current public Growth plan lists $60,000 per year plus five percent of sequencer profit and publishes adjacent RPC, indexing, and bridge entry pricing. That is useful sourcing, not an enterprise quote or total cost. PUBLIC FACT (Caldera 2026) Caldera documents OP Stack and Arbitrum Nitro deployment, but the reviewed official material does not publish a mainnet price; a current written quote is required.

The initial pilot still has no proprietary L2 and no bridge. L2 analysis becomes relevant only if measured public-chain constraints justify the additional security, governance, liquidity, operational, and exit obligations.

A.5. Five decision implications

1. **Select on total route evidence.** Cost, predictability, throughput, finality, Java/native tooling, custody, compliance, operations, maturity, strategic control, and exit risk must pass together.
2. **Keep finalities separate.** Inclusion, L1 commitment or Solana commitment, withdrawal, legal discharge, customer promise, and accounting close create different cost and risk.
3. **Do not monetize placeholder savings.** An L2 or batch benefit is not real until exact workload, failure semantics, provider terms, staffing, governance, and liquidity are measured.
4. **Preserve native engineering depth.** Low Solana fee requires first-class Rust/Anchor, account, Token-2022, CU, transaction, deployment, and monitoring capability; EVM requires equally serious contract, transaction, node, and L2-data engineering.
5. **Price the exit.** Provider portability, state and key export, bridge unwind, liquidity redemption, evidence retention, and decommissioning belong in the business case.

Pilot measurement requirement

Before chain approval, capture the exact asset and contract/program version; gas/CU/bytes/signers/accounts; base, priority, data, operator, signature and rent components; p50/p95/p99 inclusion and each required finality; failed, expired, ambiguous, retried and compensated effects; RPC/indexer and custody usage; fixed provider and staffing costs; liquidity/capital; reconciliation/support; batch behavior; and cost per completed business effect. Replay the measured workload through low/reference/high sensitivity and retain the source date and formula.

That evidence can support a profile decision. Until it exists, the numbers are a transparent discovery model, not a Zelle commitment or a network recommendation.

B. Public claim to primary source map

This appendix maps the brief's major factual topics to primary public sources. It is intentionally concise rather than sentence-by-sentence. Normal citations remain beside public facts in the narrative; the internal claim ledger retains the complete identifiers, evidence classes, source locations, caveats, and review dispositions used for regression and release validation.

Table 8. Major public claims, source coverage, and evidence limits.

Topic	Primary source(s)	Supports	Limitations
Zelle network	Zelle and Early Warning Services (Zelle 2026; Early Warning Services 2016)	Participant-bank distribution, current public scale figures, Q1 2026 survey date, and public launch context.	Does not establish internal architecture, vendors, technology, or a stablecoin and cross-border plan.
Cross-border payments	FSB remarks and World Bank data (Moloney 2026; World Bank 2025)	Cost, speed, transparency, access, data-quality, standardization, and remittance-price context.	Does not prove a Zelle-specific problem or that the proposed route improves end-to-end outcomes.
Ethereum	Ethereum Foundation documentation (Ethereum Foundation 2026b)	Inclusion, justification, and finalization terminology.	Does not establish legal, accounting, or customer finality; route performance; or selection.
Base	Base documentation (Base 2026a; Base 2026b)	L2 execution and L1 security/data fee components; L2 inclusion, L1 publication/finality, and withdrawal stages.	Does not establish fixed cost, service levels, comparative superiority, or selection.
Solana	Solana Foundation documentation (Solana Foundation 2026c; Solana Foundation 2026a; Solana Foundation 2026b)	RPC commitment levels, base and priority fees, and Rust/sBPF program execution.	Does not establish route performance, legal finality, operating fitness, custody support, or selection.
Transaction economics	Ethereum, Base, Solana, OP Stack, Nitro, Conduit, and Caldera primary sources (Ethereum Foundation 2026a; Base 2026a; Solana Foundation 2026a; Optimism Foundation 2026; Offchain Labs 2026; Conduit 2026; Caldera 2026)	Protocol fee components and current public provider entry terms used by the dated model.	Does not establish route workload, future prices, provider quotes, total cost, budget, or chain selection.
Observed Base fee	Base transaction receipt and fee documentation (BaseScan 2026; Base 2026a)	One 60-USDC transfer used 62,147 gas and paid a total fee near $0.00116 at the explorer conversion.	One dated point is not a percentile, forecast, provider cost, route cost, or customer price.
Issuer economics	Circle filings and reserve disclosures (Circle Internet Group, Inc. 2026b; Circle Internet Group, Inc. 2026a; Circle Internet Group, Inc. 2026c)	Historical reserve income, distribution cost, rate sensitivity, other revenue, and publicly described reserve composition.	Circle is a benchmark, not a Zelle forecast, margin, issuer selection, or legal/accounting conclusion.
Spring	Spring Framework documentation (Spring Team 2026)	Framework capability areas relevant to enterprise services.	Does not establish EWS adoption, production reliability, or suitability without engineering validation.
Web3j	Web3j documentation (Web3j 2026)	Java integration with Ethereum JSON-RPC and smart contracts.	Does not provide consensus, legal finality, production key controls, or evidence of organization adoption.
USDC	Circle issuer documentation (Circle 2026)	Circle's description of a multi-network issued asset and its backing statement.	Is not independent reserve assurance and does not select an asset for the pilot.
FATF guidance	Financial Action Task Force report (Financial Action Task Force 2026)	High-level illicit-finance concerns involving stablecoins, unhosted wallets, and cross-chain activity.	Does not decide a pilot's legal treatment or prove that any control stack is sufficient.
Solidity	Solidity documentation (Solidity Team 2026)	Solidity's role in EVM smart-contract development.	Does not establish that custom contracts are needed, safe, or the right place for enterprise control.

The annotated source list that follows supplies the organization, title, publication date, canonical URL, supported use, and explicit limitation for every primary source used by this release.

Annotated Primary Public Sources

Agusx1211, C. Garnett, D. Liburd, L. Schindler, et al. (2024). *ERC-7821: Minimal Batch Executor Interface*. URL: https://eips.ethereum.org/EIPS/eip-7821 (visited on 2026-07-16).
Supports: A minimal EVM batch executor interface for multiple calls in one transaction.
Does not establish: Zelle use, measured gas, approved batch size, safety, custody, or savings.

Base (2026a). *Network Fees*. URL: https://docs.base.org/base-chain/network-information/network-fees (visited on 2026-07-15).
Supports: The Base fee model's L2 execution component and L1 security or data component.
Does not establish: A fixed transaction price, total end-to-end operating cost, comparative cost superiority, or chain selection.

– (2026b). *Transaction Finality*. URL: https://docs.base.org/base-chain/network-information/transaction-finality (visited on 2026-07-15).
Supports: Distinct Base stages for L2 inclusion, L1 batch inclusion and finality, and L2-to-L1 withdrawal behavior.
Does not establish: A service-level guarantee, legal finality, route suitability, comparative performance, or chain selection.

BaseScan (2026). *Observed Base USDC Transfer Receipt*. URL: https://basescan.org/tx/0x14e19dabe9dad1a31fd7d5cda869326f0967c7469db79ca42f053c626bb5e58a (visited on 2026-07-16).
Supports: The receipt-linked amount, gas, L2 fee, L1 fee, and total fee for one dated Base USDC transfer.
Does not establish: A universal Base fee, price percentile, route cost, customer price, or forecast.

Caldera (2026). *What Are Rollups?* URL: https://docs.caldera.xyz/rollup-engine/about/about-rollups (visited on 2026-07-16).
Supports: Caldera's documented support for managed OP Stack and Arbitrum Nitro rollups.
Does not establish: A public mainnet price, enterprise quote, total cost, service fitness, or provider selection.

Circle (2026). *What is USDC?* URL: https://developers.circle.com/stablecoins/what-is-usdc (visited on 2026-07-15).
Supports: Circle's description of USDC as a multi-network issued stablecoin and its statement about liquid cash and cash-equivalent backing.
Does not establish: Independent reserve assurance, legal or accounting treatment, route availability, or selection of USDC for the pilot.

Circle Internet Group, Inc. (2026a). *Annual Report on Form 10-K for the Year Ended December 31, 2025*. URL: https://www.sec.gov/Archives/edgar/data/1876042/000187604226000062/crcl-20251231.htm (visited on 2026-07-16).
Supports: Circle's FY2025 reserve income, total revenue and reserve income, and distribution and transaction cost.
Does not establish: A ZelleUSD forecast, comparable margin, approved issuer model, or future Circle result.

– (2026b). *Quarterly Report on Form 10-Q for the Quarter Ended March 31, 2026*. URL: https://www.sec.gov/Archives/edgar/data/1876042/000187604226000150/crcl-20260331.htm (visited on 2026-07-16).
Supports: Circle's reported average USDC circulation, reserve return, reserve income, distribution costs, and rate sensitivity for Q1 2026.
Does not establish: A ZelleUSD forecast, comparable margin, approved issuer model, or future Circle result.

– (2026c). *Transparency and Stability*. URL: https://www.circle.com/transparency (visited on 2026-07-16).
Supports: Circle's description of USDC reserve asset categories and transparency reports.
Does not establish: Independent legal or accounting treatment, ZelleUSD reserve design, or selection of USDC.

Conduit (2026). *Rollup Pricing*. URL: https://www.conduit.xyz/pricing/rollups (visited on 2026-07-16).
Supports: Current public Growth plan, sequencer-profit share, and listed RPC, indexing, and bridge entry terms.
Does not establish: An enterprise quote, total cost, future pricing, contract terms, service fitness, or provider selection.

Early Warning Services (Oct. 24, 2016). *Early Warning Announces Zelle Network*. URL: https://www.zellepay.com/press-releases/early-warning-announces-zelle-network (visited on 2026-07-15).
Supports: The public 2016 announcement of the Zelle name and its bank-led distribution context.
Does not establish: Current network scale, internal implementation, present strategy, or any stablecoin and cross-border architecture.

Ethereum Foundation (2026a). *Gas and Fees*. URL: https://ethereum.org/developers/docs/gas/ (visited on 2026-07-16).
Supports: Ethereum's gas-used multiplied by base-plus-priority fee model and the 21,000-gas simple native transfer.
Does not establish: A stablecoin contract's gas use, future fee levels, route total cost, or chain selection.

– (Mar. 12, 2026b). *Transactions*. URL: https://ethereum.org/developers/docs/transactions/ (visited on 2026-07-15).
Supports: Ethereum transaction progression and the protocol terms inclusion, justification, and finalization.
Does not establish: Legal, accounting, or customer finality; route latency; a service-level guarantee; or chain selection.

Financial Action Task Force (2026). *Targeted Report on Stablecoins and Unhosted Wallets*. URL: https://www.fatf-gafi.org/en/publications/Virtualassets/targeted-report-stablecoins-unhosted-wallets.html (visited on 2026-07-15).
Supports: High-level illicit-finance concerns involving stablecoins, unhosted wallets, and cross-chain activity.
Does not establish: A legal disposition for a specific pilot, the sufficiency of a control stack, or a conclusion about Zelle.

Moloney, M. (July 8, 2026). *Cross-Border Payments: Towards the Next Chapter.* URL: https://www.fsb.org/2026/07/cross-border-payments-towards-the-next-chapter/ (visited on 2026-07-15).
Supports: The four cross-border outcomes of cost, speed, transparency, and access, plus processing friction from poor data quality and limited standardization.
Does not establish: An institutional FSB position, a Zelle-specific problem statement, or proof that a stablecoin route improves an end-to-end payment.

Offchain Labs (2026). *Arbitrum Nitro: A Second-Generation Optimistic Rollup.* URL: https://docs.arbitrum.io/nitro-whitepaper.pdf (visited on 2026-07-16).
Supports: Nitro's separation of L2 execution resources and parent-chain data cost.
Does not establish: An Orbit chain's parent, data availability, gas token, governance, provider, workload, or total cost.

Optimism Foundation (2026). *Transaction Fees.* URL: https://docs.optimism.io/op-stack/transactions/fees (visited on 2026-07-16).
Supports: The separation of OP Stack execution, L1 data, and operator fee components.
Does not establish: A future organization-specific chain configuration, price, security model, operating cost, or recommendation.

Solana Foundation (2026a). *Fees on Solana.* URL: https://solana.com/docs/core/fees (visited on 2026-07-15).
Supports: The Solana base-fee design and optional compute-unit-based prioritization fee.
Does not establish: A fixed route cost, comparative cost superiority, congestion behavior, or chain selection.

– (2026b). *Program Execution.* URL: https://solana.com/docs/core/programs/program-execution (visited on 2026-07-15).
Supports: Rust as the primary Solana program language and compilation to the chain-native sBPF execution environment.
Does not establish: The enterprise control-plane language, program safety, implementation fitness, or chain selection.

– (2026c). *Solana RPC Methods and Documentation.* URL: https://solana.com/docs/rpc (visited on 2026-07-15).
Supports: The processed, confirmed, and finalized commitment levels exposed by Solana RPC.
Does not establish: Legal or customer finality, route-specific economic safety, RPC service levels, or chain selection.

– (2026d). *Transactions.* URL: https://solana.com/docs/core/transactions (visited on 2026-07-16).
Supports: Atomic multi-instruction transactions and documented serialized-size, account, instruction, and signature constraints.
Does not establish: An approved batch size, program choice, workload cost, custody fit, or chain decision.

Solidity Team (2026). *Solidity Documentation.* URL: https://docs.soliditylang.org/en/latest/ (visited on 2026-07-15).
Supports: Solidity's role as a language for EVM smart-contract development.
Does not establish: The need for custom contracts, contract safety, enterprise orchestration ownership, or a chain decision.

Spring Team (2026). *Spring Framework Documentation.* URL: https://docs.spring.io/spring-framework/reference/index.html (visited on 2026-07-15).
Supports: Spring Framework capability areas relevant to web, messaging, transactions, persistence, integration, scheduling, caching, observability, and testing.
Does not establish: EWS or Zelle adoption, system reliability, architectural fitness without evaluation, or production performance.

United States Government Publishing Office (2026). *GENIUS Act, Compiled Public Law.* URL: https://www.govinfo.gov/content/pkg/COMPS-18221/pdf/COMPS-18221.pdf (visited on 2026-07-16).
Supports: High-level federal one-to-one reserve, restricted-use, and disclosure provisions in the compiled text.
Does not establish: Case-specific entity, product, bank, state, tax, accounting, or licensing conclusions.

Web3j (2026). *Web3j Documentation.* URL: https://docs.web3j.io/latest/ (visited on 2026-07-15).
Supports: Web3j's role as a Java library for Ethereum JSON-RPC and smart-contract integration.
Does not establish: Consensus implementation, legal finality, secure production signing, version fitness, or organization adoption.

World Bank (2025). *Remittance Prices Worldwide.* URL: https://remittanceprices.worldbank.org/ (visited on 2026-07-15).
Supports: The continued materiality of average consumer remittance cost across the corridors monitored by the World Bank database.
Does not establish: The price of every corridor, a current Zelle route, or the cost benefit of the proposed pilot.

Zelle (2026). *Innovation. Powered by Partnerships.* URL: https://www.zelle.com/join-zelle-network/partners (visited on 2026-07-16).
Supports: The participant financial-institution model, availability through more than 2,400 banking and credit-union applications, more than 143 million consumers, and the Q1 2026 survey date.
Does not establish: Internal architecture, vendors, ledger implementation, technology stack, or a cross-border and stablecoin plan.